

# Who gains from a market upgrade? Stock liquidity and prices around Vietnam's FTSE Russell reclassification

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## Abstract

FTSE Russell reclassified Vietnam from frontier to secondary emerging status in four public steps between October 2025 and September 2026. Using daily data on 366 stocks listed on the Ho Chi Minh City Stock Exchange, we trace liquidity and prices at each step. Because FTSE chose the final constituents on mid-2026 data, we estimate effects both for the 24 eventual constituents and, as an intention-to-treat design, for the 27 stocks FTSE had screened as eligible on data from before the announcement. Relative to never-named stocks, the Amihud illiquidity of the pre-announcement eligible group fell by 24%, 43% and 56% after the announcement, the confirmation and the constituent list; for the constituents the declines were 32%, 59% and 68%. Constituents earned cumulative abnormal returns of 3.9% to 7.8% in the week around the announcement and 2.7% to 4.5% around the confirmation, significant under four benchmarks and a test that allows for the common event dates; the constituent list and the effective date of the first 10% index tranche produced no reliable price effect. Trading value surged at the rebalancing close for included stocks but not for stocks FTSE named and then excluded. The liquidity and price response to Vietnam's upgrade came with FTSE's disclosures, before the first index tranche took effect.

**Keywords:** market reclassification, index effect, stock liquidity, frontier markets, intention to treat, Vietnam

## **1. Introduction**

Vietnam's market upgrade raised the liquidity of likely index stocks before the first index tranche took effect. When FTSE Russell reclassified Vietnam from frontier to secondary emerging status, the Amihud (2002) illiquidity of the 27 stocks FTSE had screened as eligible on 2024 data fell relative to never-named stocks on the Ho Chi Minh City Stock Exchange (HOSE) by 43% after FTSE confirmed the upgrade in April 2026 and by 56% after it named the constituents in August 2026. For the stocks FTSE finally included, the declines were 59% and 68%. On the effective date, which carried the first 10% of the eventual index weight, prices did not move; trading value surged at the rebalancing close for included stocks only.

Market classification decides which benchmarks a country's stocks can enter, and benchmark weights steer international portfolio allocations (Raddatz et al., 2017). An upgrade from frontier to emerging status therefore promises new foreign demand, and Vietnamese policymakers pursued it through seven years of watch-list reviews. Whether the promised liquidity reaches individual stocks, which stocks it reaches, and when, bears on issuers' cost of capital, because liquidity is priced in emerging markets (Bekaert et al., 2007).

Existing evidence answers these questions poorly for frontier markets. In developed markets, additions to the S&P 500 raise trading activity and narrow spreads persistently (Hegde & McDermott, 2003), and added firms with larger liquidity gains invest more (Becker-Blease & Paul, 2006). In frontier markets,

index additions raise prices persistently, and the evidence points to institutional demand rather than trading pressure or liquidity changes as the cause (Biktimirov & Afego, 2026). Most studies of market reclassification also work with aggregate indices or country-level flows (e.g., Burnham et al., 2018; Raddatz et al., 2017), which cannot separate the stocks an upgrade targets from market-wide trends. An earlier study of the same market by the present authors (anonymized for review) found that FTSE watch-list reviews did not coincide with structural breaks in index-level liquidity; that design could not ask which stocks gained.

Vietnam's reclassification offers a cleaner test for three reasons. FTSE Russell disclosed the decision in four dated steps: the announcement on 7 October 2025, the confirmation on 7 April 2026, the constituent list on 21 August 2026, and the effective date of 21 September 2026. Each step resolved a different piece of uncertainty, so their separate effects reveal when investors acted. Hundreds of HOSE stocks outside the index trade under the same rules, hours and price limits and provide a comparison group. And FTSE published lists of eligible stocks before it chose the constituents, including a preliminary list screened on data as of 31 December 2024, before the upgrade was announced. That list lets us estimate intention-to-treat effects that do not depend on FTSE's later choice.

The last point matters because FTSE selected the final constituents at its September 2026 review, whose liquidity screen uses data up to the last business day of June 2026 (FTSE Russell, n.d.), in the middle of the period we study. A stock whose trading grew after October 2025 for any reason was more likely to be selected, so a comparison of final constituents with other stocks could mistake

selection for an upgrade effect. We address this in two ways. We remove from the comparison group every stock FTSE named as eligible but did not include, and we estimate intention-to-treat effects for the preliminary list.

We estimate difference-in-differences and event-study regressions on 35,752 stock-weeks and on daily returns from October 2024 to September 2026. The main liquidity measure is the Amihud ratio computed on traded value in Vietnamese dong; we also report trading value, the Corwin and Schultz (2012) high-low spread, volatility and abnormal returns. For prices, we pre-specify the same three windows for every disclosure, test cumulative abnormal returns at the portfolio level, which accounts for the common event dates (Brown & Warner, 1985), and call an effect reliable only if it is significant at 5% under all four benchmarks we use.

The paper makes one contribution: it decomposes the stock-level liquidity and price effects of a frontier-to-emerging reclassification by disclosure stage, with the treatment group also defined on information available before the upgrade. Four findings support the decomposition. First, the illiquidity of the pre-announcement eligible group fell by 0.27, 0.55 and 0.81 log points across the announcement, confirmation and constituent-list windows, about two thirds of the 0.39, 0.89 and 1.13 log-point declines for the final constituents (Tables 2 and 3); the confirmation brought a discrete step beyond a gradual post-announcement drift, the constituent list did not. Second, prices rose reliably in the announcement week and at the confirmation, and the confirmation gain persisted under three of four benchmarks; the constituent list and the effective date produced no reliable price effect (Table 4). Third, trading value rose by 0.79 log points relative to never-named stocks at

the rebalancing close, rising with FTSE size segment, while named-but-excluded stocks showed no surge (Tables 5 and 6). Fourth, FTSE's eligibility lists partly followed rising liquidity: GEE, added in April 2026, had become more liquid during the data window FTSE screened (Table 9). This last result limits causal readings of the constituent estimates and is why we report intention-to-treat effects beside them.

Section 2 describes the setting, reviews the literature and states the hypotheses. Section 3 presents the data and design. Section 4 reports the results, Section 5 the robustness checks and the selection evidence, and Section 6 concludes.

## **2. Setting, literature and hypotheses**

### *2.1 FTSE Russell's four-step reclassification of Vietnam*

FTSE Russell added Vietnam to its watch list for possible reclassification in September 2018 (FTSE Russell, 2018). On 7 October 2025 it announced that Vietnam would move from frontier to secondary emerging status with effect from 21 September 2026, subject to an interim review in March 2026 (LSEG, 2025). On 7 April 2026 it confirmed that Vietnam met all criteria for secondary emerging status and kept the effective date (LSEG, 2026). Inclusion is phased in four tranches that add 10%, 20%, 35% and 35% of the applicable index weight on 21 September 2026, 22 March 2027, 21 June 2027 and 20 September 2027 (FTSE Russell, 2026). The effective date we study therefore carried only the first tenth of the eventual index weight.

Between these dates the financial press reported FTSE's lists of eligible stocks.

A preliminary list of 28 names appeared in November 2025 (Viet Nam News, 2025); it was screened on data as of 31 December 2024 (The Investor, 2026a). A list of 32 names, screened on data as of 31 December 2025, followed in April 2026; it dropped Petrolimex (PLX) and added BID, FPT, NVL, GEE and BSR (The Investor, 2026b). On Friday 21 August 2026 FTSE published the September semi-annual review with 27 constituents, effective after the close of 18 September 2026 (FTSE Russell, 2026; VIR, 2026); under FTSE's ground rules the September review screens liquidity on data up to the last business day of June (FTSE Russell, n.d.). FTSE screened Vietnamese securities as non-constituents on its liquidity, minimum-size and foreign-headroom screens, using each security's investability weight, which reflects its free float and foreign-ownership limits, in the liquidity test (FTSE Russell, 2026). The same investability weight scales a security's weight in the index, and each tranche applies a fraction of it: a security with a 49% investability weight enters at 4.9% after the first tranche (10% of 49%) and at 49% after the last. A constituent's index weight, and the passive demand it attracts, therefore rises with its free-float market capitalization. Appendix Table A.2 lists every stock by list.

The four steps carried different information. The announcement told investors that Vietnamese stocks would become eligible for emerging-market benchmarks but made the upgrade conditional on an interim review. The confirmation removed that condition. The constituent list named the stocks index-tracking funds would buy, and the rebalancing session was the day they bought the first tranche. Effects that appear at the early disclosures indicate that investors traded on information;

effects concentrated at the rebalancing session indicate mechanical index demand.

## *2.2 Index inclusion and liquidity*

Two channels link index membership to liquidity and prices. The price-pressure channel works through the trades of index funds: additions lead index funds to buy, which produces temporary volume and price effects that reverse afterwards (Harris & Gurel, 1986). The demand and recognition channel works through a lasting change in the investor base. Shleifer (1986) interprets excess returns that persist for at least ten days after inclusion as evidence that demand curves for stocks slope down, and Chen et al. (2004) find asymmetric effects of additions and deletions that fit a gain in investor awareness. Hegde and McDermott (2003) show that S&P 500 additions narrow spreads and raise trading activity persistently, which they attribute mainly to lower direct trading costs.

Liquidity gains from inclusion also have real consequences. Becker-Blease and Paul (2006) find that firms added to the S&P 500 with larger liquidity improvements increase capital investment, whereas Gregoriou and Nguyen (2010) find that the liquidity lost after FTSE 100 deletions has no significant effect on investment. Liquidity co-moves across stocks (Chordia et al., 2000), so any study of these effects must separate stock-specific changes from market-wide liquidity shocks.

## *2.3 Market reclassification and frontier markets*

Reclassification moves a whole market between benchmark families. Mutual funds allocate across countries with reference to benchmark weights, so a change

in classification changes capital flows (Raddatz et al., 2017), and an MSCI index change even moved currency values through this channel (Hau et al., 2010). Burnham et al. (2018) show that, when MSCI moves a country to a different index family, the country's stock prices move sharply in the direction implied by the new benchmark before the change is implemented and give back most of that move in the following year. When MSCI included China A-shares in its Emerging Markets Index, the included stocks earned abnormal returns around the announcement, and market quality changed over the longer run through liquidity, turnover and price synchronization (Dong et al., 2023). The frontier-market evidence is thinner. Biktimirov and Afego (2026) study changes to the FTSE Frontier 50 Index from 2008 to 2025. They find persistent price gains for additions and separate return and trading-volume responses when a country changes class, and they trace the gains to institutional demand rather than to trading pressure or liquidity. Our design complements theirs: it follows liquidity stage by stage and compares index stocks with non-index stocks of the same market. No study we know of, in a search bounded to Crossref records and the literature cited here, estimates the stock-level liquidity effect of a frontier-to-emerging reclassification with a comparison group and a pre-determined treatment list.

Liquidity carries particular weight in these markets. Bekaert et al. (2007) show that local market liquidity predicts returns in emerging markets, where segmentation from global capital is only partial, and Stereńczak et al. (2020) ask whether frontier-market investors demand compensation for illiquidity. A reclassification that improves liquidity for a subset of stocks could therefore shift their cost of capi-

tal relative to the rest of the market; we do not measure that shift here. Frontier markets also differ from developed ones in who trades: when domestic retail investors dominate turnover, the arrival of benchmark-tracking foreign money is a larger change in the investor base than an S&P 500 addition. The Vietnamese evidence available before the upgrade concerns market-wide liquidity shocks rather than classification events; for example, Nguyen et al. (2021) study the returns and liquidity of financial-services stocks during the COVID-19 outbreak. Beyond Dong et al. (2023), we did not locate a verified stock-level study of liquidity around the MSCI inclusion of China A-shares, so our comparison with that episode rests on their evidence alone.

#### *2.4 Hypotheses*

The recognition channel predicts that liquidity and prices respond when information reaches investors; the price-pressure channel predicts a trading and price spike when index funds rebalance, followed by reversal.

H1. After the reclassification announcement, stocks likely to enter the index became less illiquid than other HOSE stocks.

H2. The liquidity gap widened at each later disclosure that resolved uncertainty about the upgrade and its constituents.

H3. Likely constituents earned abnormal returns at the disclosures, and the first index tranche added no further price effect.

H4. The rebalancing session produced a trading-value surge for constituents, and not for excluded stocks, that scaled with index weight.

### 3. Data and empirical design

#### 3.1 Sample and groups

We collect daily open, high, low and close prices and share volume for all 405 common stocks listed on HOSE as of 24 September 2026, from 1 October 2024 to 23 September 2026. The data come from the VCI feed through the open-source vnstock library. We keep stocks with at least 200 trading days before 7 October 2025, zero volume on fewer than 20% of those days, and a median price of at least VND 1,000, which leaves 366 stocks.

We sort them into three groups using FTSE’s public lists (Appendix Table A.2). Constituents are the 24 stocks of the final 27 with a full pre-period; TCX, VPL and VCK listed after October 2025. Named-but-excluded stocks are the 14 HOSE stocks that appeared on the November 2025 or April 2026 eligible list but not among the constituents (BSR, DGC, DIG, DPM, DXG, EIB, FRT, GEE, KBC, KDC, KDH, PDR, PLX and SAB); the fifteenth such name, Tasco (HUT), trades on the Hanoi exchange. The comparison group consists of the remaining 328 never-named stocks. The intention-to-treat (ITT) group consists of the 27 HOSE stocks on the preliminary list screened on 2024 data, of which 15 later became constituents and 12 did not. BSR, one of the named-but-excluded stocks, traded on the UPCoM market until 6 January 2025 and on HOSE from 17 January 2025 (VietnamPlus, 2024), so its pre-period series spans two venues; Section 5.2 reports results without it.

Table 1 shows that constituents are much larger and more liquid than the average HOSE stock. Before the announcement, constituents traded a mean of 6.82 log

VND billion per week against 2.19 for never-named stocks, and their mean weekly Amihud ratio was about three orders of magnitude lower. Named-but-excluded stocks sit between the two. Spreads and absolute returns are similar across groups. The size gap motivates the matched sample and the size-specific checks below.

**Table 1. Sample composition and pre-announcement characteristics (October 2024 to 6 October 2025)**

| Group                                 | Stocks | Stock-weeks | Mean weekly Amihud | Mean log trading value (VND bn) | Mean CS spread (%) | Mean absolute return (%) |
|---------------------------------------|--------|-------------|--------------------|---------------------------------|--------------------|--------------------------|
| FTSE constituents                     | 24     | 1,248       | 0.00017            | 6.82                            | 0.57               | 1.40                     |
| Named on eligible lists, not included | 14     | 726         | 0.0033             | 5.96                            | 0.61               | 1.49                     |
| Never-named HOSE stocks               | 328    | 16,915      | 0.195              | 2.19                            | 0.61               | 1.37                     |

*Notes:* Amihud is the mean daily ratio of absolute log return to traded value (VND billion) within a week. CS is the Corwin and Schultz (2012) spread. Source: `output/revision/t1_descriptives.csv`.

### 3.2 Liquidity and return measures

Daily traded value equals the closing price times share volume, in VND billion. Following Amihud (2002), who divides the absolute daily return by dollar volume, we define illiquidity for stock  $i$  on day  $t$  with the log return and traded value in VND as

$$\text{ILLIQ}_{it} = \frac{|R_{it}|}{\text{VAL}_{it}},$$

where  $R_{it}$  is the log return and  $VAL_{it}$  the traded value. Traded value, rather than share volume, makes the ratio comparable across stocks with different price levels. We compute the Corwin and Schultz (2012) spread from two-day high and low prices and set negative estimates to zero. We winsorize the daily Amihud ratio and spread cross-sectionally at the 1st and 99th percentiles of each day and aggregate to a stock-week panel, keeping weeks with at least three trading days and a positive Amihud ratio. Weekly Amihud is the mean of daily ratios, used in logs; weekly trading value is the log of total traded value; the weekly spread is the mean daily spread; weekly volatility is the log of the mean absolute daily return. Kang and Zhang (2014) show that the Amihud ratio loses accuracy in emerging markets where many days have zero volume, and propose an adjustment; our filter removes stocks with zero volume on 20% or more of pre-period days, which limits this problem.

For prices, the daily abnormal return of stock  $i$  is its log return minus a benchmark return. We use four benchmarks: the equal-weighted mean of the 328 never-named stocks; the mean of the matched control stocks described below, weighted by their matching weights; the equal-weighted mean of the 110 never-named stocks in the top tercile of pre-period trading value among never-named stocks; and a market model with intercept and slope estimated for each stock against the first benchmark. Because every constituent shares the same event dates, abnormal returns are correlated across stocks, and cross-sectional t-statistics overstate precision. We therefore test cumulative abnormal returns (CARs) on the equal-weighted portfolio of the group, which addresses cross-sectional dependence (Brown & Warner,

1985): the test statistic divides the portfolio CAR by the product of two terms, the standard deviation of daily portfolio abnormal returns in an estimation window and the square root of the number of days in the event window, which assumes independent daily abnormal returns. The estimation window runs from trading day -130 to -11 before each event and excludes days -1 to +20 around every earlier disclosure, which removes 22 to 31 event days from the windows of the later disclosures. We pre-specify three windows for every disclosure, [-1,+1], [-1,+5] and [0,+20], report all of them, and call a CAR reliable when it is significant at 5% under all four benchmarks.

### 3.3 Estimation

The baseline liquidity specification is

$$y_{iw} = \alpha_i + \lambda_w + \sum_{k=1}^3 \beta_k (\text{Constituent}_i \times P_{kw}) + \varepsilon_{iw},$$

where  $y_{iw}$  is a liquidity measure for stock  $i$  in week  $w$ ,  $\alpha_i$  and  $\lambda_w$  are stock and week fixed effects, and  $P_{1w}$ ,  $P_{2w}$  and  $P_{3w}$  indicate the announcement window (7 October 2025 to 6 April 2026), the confirmation window (7 April to 20 August 2026) and the list-and-rebalancing window (21 August to 23 September 2026). Weeks start on Monday, and a week enters a window only if it starts on or after the disclosure date, so each window begins with the first full week after the disclosure. FTSE does not state the hour at which it released the constituent list; the weekly assignment and the [-1,+1] event window, which spans 20, 21 and 24 August, both contain the first Vietnamese session after the release. The baseline sample excludes

named-but-excluded stocks, so the comparison group contains only never-named stocks. Week fixed effects absorb market-wide shocks. We cluster standard errors by stock.

The ITT specification replaces  $\text{Constituent}_i$  with an indicator for the 27 stocks on the preliminary list, keeps every other stock in the sample, and in a variant restricts the comparison group to never-named, never-included stocks. A third treatment, “predicted constituents”, takes the 27 stocks with the highest pre-announcement trading value; 16 of them became constituents.

Constituents are the largest HOSE stocks, so shocks that affect large stocks differently could bias the comparison. We re-estimate on a matched sample that pairs each constituent with three never-named stocks by nearest-neighbour propensity score, with replacement, on pre-period traded value, price and return volatility (Ho et al., 2007); because matching is with replacement, each control enters the estimates with its matching weight. Matching reduces the standardized mean difference in pre-period traded value from 4.06 to 0.04 (Appendix Table A.1). We also replace week fixed effects with size-tercile-by-week fixed effects, add a constituent-specific linear trend, and estimate a monthly event study relative to September 2025 (Roth et al., 2023). For inference with 24 treated stocks, we report a wild cluster bootstrap (Cameron et al., 2008) and randomization inference that draws 500 pseudo-treated groups of 24 stocks from the 85 never-named stocks whose pre-period trading value lies in the top tercile of all 366 sample stocks. Because FTSE treated all constituents at the same dates, the specification has no staggered adoption, and the two-way fixed effects estimator does not suffer the

negative-weighting problem that motivates Callaway and Sant’Anna (2021).

For the rebalancing session we estimate a daily regression of log traded value on stock and date fixed effects and constituent indicators for 18 September (the rebalancing session) and 21 September (the effective date), over the 32 sessions from 6 August to 23 September 2026. The coefficients measure the surge on those sessions relative to the other sessions in the window.

### *3.4 Identification assumptions*

The coefficients  $\beta_k$  identify the effect of the upgrade on constituents under four assumptions.

First, parallel trends: without the upgrade, constituents and never-named stocks would have followed the same liquidity path. The event study tests its observable implication for the pre-announcement months, and the matched, size-by-week and trend specifications relax it in different directions.

Second, no anticipation before 7 October 2025. Vietnam sat on the FTSE watch list for seven years (FTSE Russell, 2018; LSEG, 2025), so investors could have bought likely constituents in advance of the decision. Anticipation of this kind would raise liquidity before the announcement and bias the estimates toward zero.

Third, no spillovers onto comparison stocks. If investors sold never-named stocks to buy constituents, the comparison group’s liquidity would fall because of the treatment. Removing named-but-excluded stocks from the comparison group, as we do, prevents the most likely spillover channel, trading in near-substitutes, from contaminating it.

Fourth, the treatment group must not be selected on post-treatment outcomes. This assumption fails for the final constituents by construction, because FTSE selected them on data as of 30 June 2026. The ITT group avoids the problem: FTSE screened it on data from 2024. ITT estimates measure the average effect on stocks that were likely constituents before the announcement, including the 12 listed stocks that FTSE did not include.

## **4. Results**

### *4.1 Likely constituents became less illiquid from the announcement onward*

Constituents' Amihud illiquidity fell relative to never-named stocks in each disclosure window, and the fall deepened at each stage. Table 2, column 1, reports coefficients of -0.39, -0.89 and -1.13 log points, declines of 32%, 59% and 68%. All three are significant at the 0.1% level, and the wild cluster bootstrap p-values are below 0.001. The matched sample, with each control weighted by its matching weight, gives smaller declines: -0.23 (not significant), -0.47 and -0.69 log points in column 4 (declines of 21%, 38% and 50%). The confirmation brought a discrete step: allowing constituents a linear drift after the announcement (-0.012 log points per week,  $p = 0.002$ ), the confirmation-window coefficient still exceeds the announcement-window coefficient by 0.21 log points (standard error 0.10,  $p = 0.035$ ). The constituent list did not: its additional step is 0.10 (standard error 0.11,  $p = 0.40$ ). H2 therefore holds for the confirmation; after it, liquidity kept improving gradually rather than jumping at the list.

**Table 2. Liquidity of FTSE constituents relative to never-named HOSE stocks, by disclosure stage**

|                                          | (1) log<br>Amihud           | (2) log<br>trading<br>value | (3) CS<br>spread            | (4) log<br>Amihud           | (5) log<br>trading<br>value | (6) CS<br>spread            |
|------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| Sample                                   | All never-named<br>controls | All never-named<br>controls | All never-named<br>controls | Matched<br>3:1,<br>weighted | Matched<br>3:1,<br>weighted | Matched<br>3:1,<br>weighted |
| Constituent ×<br>Announcement            | -0.392***<br>(0.098)        | 0.678***<br>(0.108)         | 0.0009**<br>(0.0003)        | -0.233<br>(0.159)           | 0.387*<br>(0.166)           | 0.0011**<br>(0.0004)        |
| Constituent ×<br>Confirmation            | -0.888***<br>(0.124)        | 1.030***<br>(0.125)         | 0.0010**<br>(0.0003)        | -0.471**<br>(0.167)         | 0.642***<br>(0.177)         | 0.0014***<br>(0.0004)       |
| Constituent ×<br>List and<br>rebalancing | -1.129***<br>(0.178)        | 1.189***<br>(0.168)         | 0.0013**<br>(0.0005)        | -0.691**<br>(0.232)         | 0.845**<br>(0.245)          | 0.0013*<br>(0.0006)         |
| Stock and week<br>FE                     | Yes                         | Yes                         | Yes                         | Yes                         | Yes                         | Yes                         |
| Observations                             | 34,369                      | 34,369                      | 34,369                      | 4,751                       | 4,751                       | 4,751                       |

*Notes:* Stock-week panel, October 2024 to September 2026; named-but-excluded stocks removed. Standard errors clustered by stock in parentheses. \*, \*\*, \*\*\* denote significance at 5%, 1% and 0.1%. Columns 4 to 6 weight each of the 24 distinct matched controls by its matching weight (three controls per constituent, nearest-neighbour propensity score, with replacement). Sources: `output/revision/t2_baseline_clean.csv`, `output/revision2/t_matched_weighted.csv`.

The ITT estimates in Table 3 indicate that the decline is not an artefact of FTSE’s later choice. The 27 stocks screened on 2024 data became less illiquid by 0.27, 0.55 and 0.81 log points (declines of 24%, 43% and 56%), all significant at the 1% level or better, and slightly more when the comparison group excludes every named or included stock. The ITT effects are 62% to 72% of the constituent

effects, as expected when 12 of the 27 listed stocks were never included. Splitting the ITT group shows where the gains concentrated: listed stocks that later entered the index gained 0.43, 0.87 and 1.12 log points, while listed stocks that did not enter gained 0.11, 0.25 and 0.53, the last significant at 5%. The 27 predicted constituents, chosen by pre-period trading value alone, gained 0.25, 0.59 and 0.74 log points.

**Table 3. Intention-to-treat and predicted-constituent estimates (log Amihud unless stated)**

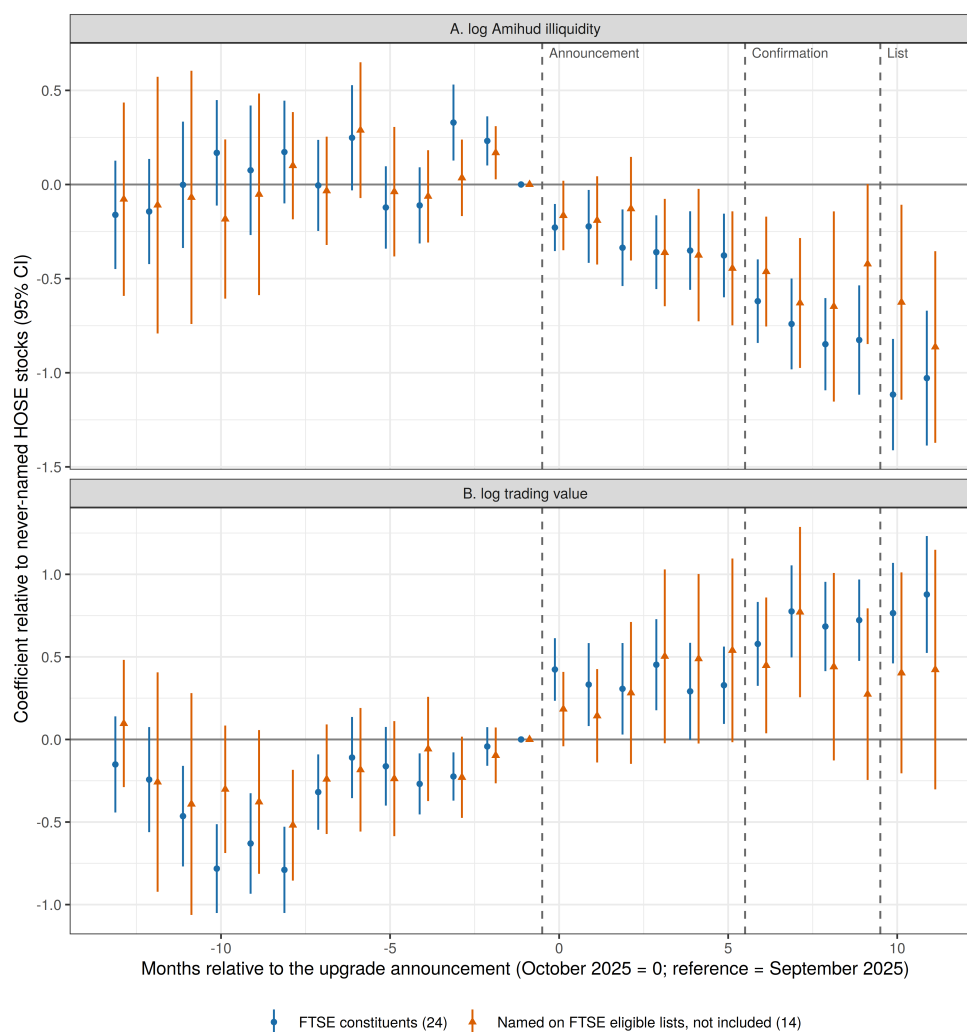
| Treatment group                                                  | Announcement         | Confirmation         | List and rebalancing | log trading value, list window | Observations |
|------------------------------------------------------------------|----------------------|----------------------|----------------------|--------------------------------|--------------|
| ITT: preliminary list (27), all other stocks as controls         | -0.268**<br>(0.083)  | -0.554***<br>(0.129) | -0.811***<br>(0.167) | 0.724***<br>(0.194)            | 35,752       |
| ITT: preliminary list (27), never-named, never-included controls | -0.286***<br>(0.083) | -0.593***<br>(0.129) | -0.860***<br>(0.167) | 0.776***<br>(0.194)            | 34,666       |
| of which later included (15)                                     | -0.428***<br>(0.101) | -0.872***<br>(0.146) | -1.123***<br>(0.192) |                                | 34,666       |
| of which not included (12)                                       | -0.108<br>(0.102)    | -0.245<br>(0.159)    | -0.527*<br>(0.236)   |                                | 34,666       |
| Predicted constituents: top 27 by pre-period trading value       | -0.250***<br>(0.073) | -0.595***<br>(0.089) | -0.741***<br>(0.120) | 0.607***<br>(0.144)            | 35,752       |

*Notes:* The preliminary list is FTSE's November 2025 eligible list screened on data as of 31 December 2024 (27 HOSE stocks). Stock and week fixed effects; standard errors clustered by stock. The split rows come from one regression with separate interactions. Source: output/revision/t\_itt.csv.

Figure 1 shows the timing. The constituents' monthly Amihud coefficients move around zero in the pre-announcement year without a downward drift. They are positive in several months, most clearly three and two months before the announcement (0.33 and 0.23), so constituents were, if anything, becoming less liquid just before October 2025. Because these deviations are precisely estimated, the joint test that all pre-period coefficients are zero rejects in the full sample ( $F = 6.66$ ,  $p < 0.001$ ); in the matched sample it does not reject at 5% ( $F = 1.72$ ,  $p =$

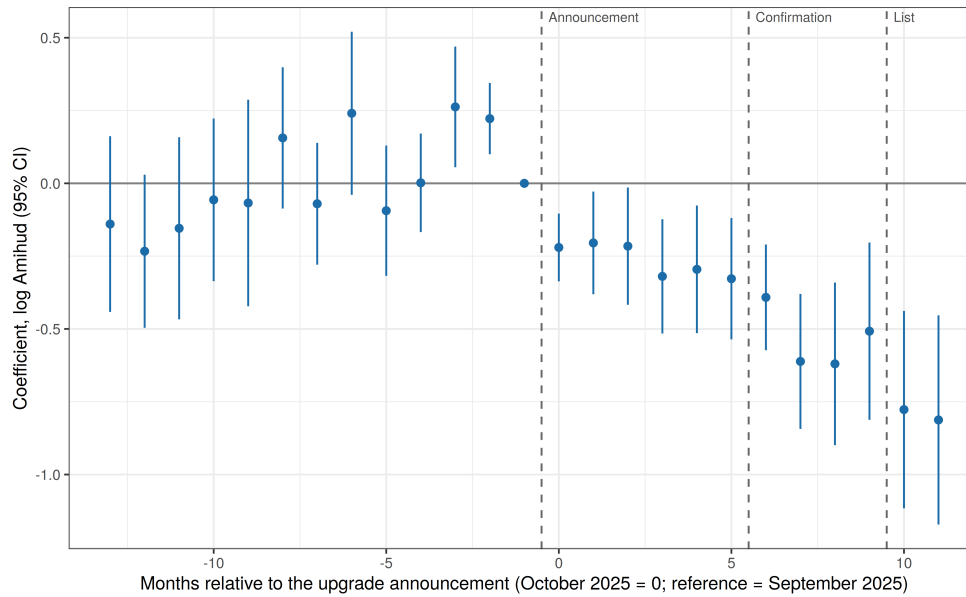
0.092). These p-values use the number of stock clusters minus one as denominator degrees of freedom. The coefficient turns negative in October 2025 (-0.23), steps down after the confirmation (-0.62 in April 2026) and reaches -1.12 in August 2026. Section 5 shows that the estimates do not depend on the months before the announcement. The ITT group shows the same shape (Figure 2): coefficients of 0.26 and 0.22 three and two months before the announcement, which make its joint pre-trend test reject as well ( $F = 7.59$ ,  $p < 0.001$ ), followed by a decline from -0.22 in October 2025 to -0.78 in August 2026.

**Figure 1. Monthly event-study coefficients for FTSE constituents and named-but-excluded stocks**



*Note:* Coefficients relative to September 2025 from one regression per outcome with stock and week fixed effects; the comparison group is never-named HOSE stocks. 95% confidence intervals from stock-clustered standard errors. Dashed lines mark the announcement, confirmation and constituent-list months. Source: `output/revision/t_event_study.csv`.

**Figure 2. Monthly event-study coefficients for the intention-to-treat group**



*Note:* log Amihud illiquidity of the 27 stocks on FTSE’s preliminary eligible list (screened on 2024 data) relative to never-named, never-included HOSE stocks; coefficients relative to September 2025 from a regression with stock and week fixed effects; 95% confidence intervals from stock-clustered standard errors. Source: output/revision2/t\_itt\_event\_study.csv.

Trading value tells a similar story with one caveat. Constituents’ trading value rose by 0.68, 1.03 and 1.19 log points relative to never-named stocks (Table 2, column 2), but it was already rising before October 2025 (Figure 1, panel B). Trend-adjusted effects fall to 0.28, 0.39 and 0.43 log points (Section 5), and we describe the trading-value result as an association.

Randomization inference confirms that the declines are not what large HOSE stocks experienced in general. None of 500 pseudo-groups of 24 stocks drawn from the 85 largest never-named stocks produced a coefficient as negative as the

observed one in any window (one-sided  $p < 0.002$ ). The placebo distribution centres at -0.21 in the list window: large stocks became somewhat more liquid, by less than one fifth of the constituent effect.

**Takeaway.** Stocks that were likely constituents before the announcement became about one quarter less illiquid within months of it and more than half less illiquid by the constituent list; the stocks FTSE finally included gained most.

#### *4.2 Prices moved at the announcement and the confirmation*

Constituents earned abnormal returns in the week around the announcement and around the confirmation, and these are the only disclosure-window CARs that are reliable in the sense defined in Section 3.2. Table 4, panel A, reports CARs with portfolio t-statistics for every pre-specified window. Over days -1 to +5 around the announcement, constituents earned between 3.9% and 7.8% depending on the benchmark (t between 2.19 and 3.46). Over days -1 to +1 around the confirmation, they earned between 2.7% and 4.5% (t between 2.22 and 2.93). The three-day announcement window is significant under one benchmark only, so the announcement response built up over the week. Around the constituent list, three-day CARs of 1.5% to 2.7% are not significant under any benchmark, and the [-1,+5] CARs of 2.0% to 4.4% are significant under two of four. Around the effective date the three-day CAR lies between -0.5% and 0.9% and is never significant. The last column shows why the portfolio test matters: cross-sectional t-statistics, which ignore the common event date, are up to 2.9 times larger.

**Table 4. Cumulative abnormal returns around FTSE disclosures: portfolio tests**

Panel A. FTSE constituents (24): CAR (portfolio t) by benchmark

| Event                         | Window  | Never-named, equal-weighted | Matched controls, weighted | Large never-named stocks | Market model     | Cross-sectional t (first benchmark) |
|-------------------------------|---------|-----------------------------|----------------------------|--------------------------|------------------|-------------------------------------|
| Announcement, 7 Oct 2025      | [-1,+1] | 3.3%<br>(2.27)              | 1.7%<br>(1.43)             | 2.3%<br>(1.54)           | 2.6%<br>(1.85)   | 6.48                                |
|                               | [-1,+5] | 7.8%<br>(3.46)              | 3.9%<br>(2.19)             | 6.3%<br>(2.75)           | 6.6%<br>(3.04)   | 5.18                                |
|                               | [0,+20] | -2.4%<br>(-0.61)            | 1.6%<br>(0.51)             | -1.4%<br>(-0.35)         | -5.5%<br>(-1.47) | -0.80                               |
| Confirmation, 7 Apr 2026      | [-1,+1] | 4.5%<br>(2.93)              | 2.7%<br>(2.22)             | 3.6%<br>(2.73)           | 3.8%<br>(2.73)   | 7.50                                |
|                               | [-1,+5] | 4.7%<br>(2.00)              | 2.2%<br>(1.21)             | 3.5%<br>(1.75)           | 3.6%<br>(1.72)   | 4.81                                |
|                               | [0,+20] | 10.0%<br>(2.44)             | 6.2%<br>(1.94)             | 10.6%<br>(3.03)          | 8.4%<br>(2.30)   | 3.68                                |
| Constituent list, 21 Aug 2026 | [-1,+1] | 2.7%<br>(1.86)              | 1.5%<br>(1.31)             | 1.8%<br>(1.50)           | 1.5%<br>(1.25)   | 5.23                                |
|                               | [-1,+5] | 4.4%<br>(2.04)              | 2.0%<br>(1.12)             | 3.9%<br>(2.17)           | 3.1%<br>(1.64)   | 5.15                                |
|                               | [0,+20] | 4.6%<br>(1.21)              | 3.7%<br>(1.21)             | 5.2%<br>(1.65)           | 3.3%<br>(1.00)   | 2.87                                |
| Effective date, 21 Sep 2026   | [-1,+1] | -0.1%<br>(-0.11)            | 0.9%<br>(0.88)             | 0.0%<br>(0.03)           | -0.5%<br>(-0.51) | -0.20                               |

Panel B. Intention-to-treat group (27 stocks on the preliminary list): CAR (portfolio t) by benchmark

| Event                    | Window  | Never-named, equal-weighted | Matched controls, weighted | Large never-named stocks | Market model |
|--------------------------|---------|-----------------------------|----------------------------|--------------------------|--------------|
| Announcement, 7 Oct 2025 | [-1,+1] | 3.0% (2.43)                 | 1.3% (1.07)                | 2.0% (1.87)              | 2.3% (2.05)  |

| Event                            | Window  | Never-named,<br>equal-weighted | Matched<br>controls,<br>weighted | Large<br>never-named<br>stocks | Market<br>model  |
|----------------------------------|---------|--------------------------------|----------------------------------|--------------------------------|------------------|
| Confirmation, 7 Apr<br>2026      | [-1,+5] | 7.5% (3.97)                    | 3.5% (1.91)                      | 5.9% (3.72)                    | 6.4% (3.75)      |
|                                  | [0,+20] | -1.0%<br>(-0.32)               | 2.9% (0.90)                      | -0.1%<br>(-0.03)               | -3.6%<br>(-1.20) |
|                                  | [-1,+1] | 3.1% (2.17)                    | 1.3% (0.99)                      | 2.2% (2.10)                    | 2.4% (2.02)      |
|                                  | [-1,+5] | 3.2% (1.45)                    | 0.7% (0.35)                      | 2.0% (1.24)                    | 2.3% (1.25)      |
| Constituent list, 21<br>Aug 2026 | [0,+20] | 7.0% (1.83)                    | 3.2% (0.94)                      | 7.6% (2.74)                    | 6.3% (2.01)      |
|                                  | [-1,+1] | 2.7% (2.08)                    | 1.6% (1.37)                      | 1.8% (2.13)                    | 1.7% (1.64)      |
|                                  | [-1,+5] | 3.2% (1.62)                    | 0.7% (0.42)                      | 2.7% (2.05)                    | 2.3% (1.43)      |
| Effective date, 21<br>Sep 2026   | [0,+20] | 1.2% (0.35)                    | 0.3% (0.11)                      | 1.8% (0.80)                    | 1.4% (0.50)      |
|                                  | [-1,+1] | 0.4% (0.35)                    | 1.5% (1.52)                      | 0.6% (0.75)                    | 0.2% (0.27)      |
|                                  |         |                                |                                  |                                |                  |

Panel C. Named-but-excluded stocks (14): CAR (portfolio t) by benchmark

| Event                            | Window  | Never-named,<br>equal-weighted | Matched<br>controls,<br>weighted | Large<br>never-named<br>stocks | Market<br>model  |
|----------------------------------|---------|--------------------------------|----------------------------------|--------------------------------|------------------|
| Announcement, 7<br>Oct 2025      | [-1,+1] | 1.5% (1.11)                    | -0.1%<br>(-0.09)                 | 0.5% (0.50)                    | 1.1% (0.87)      |
|                                  | [-1,+5] | 6.2% (2.97)                    | 2.3% (0.94)                      | 4.7% (3.10)                    | 5.9% (3.14)      |
|                                  | [0,+20] | 4.8% (1.32)                    | 8.7% (2.04)                      | 5.8% (2.18)                    | 4.5% (1.39)      |
| Confirmation, 7 Apr<br>2026      | [-1,+1] | 0.4% (0.26)                    | -1.4%<br>(-0.75)                 | -0.5%<br>(-0.41)               | -0.4%<br>(-0.29) |
|                                  | [-1,+5] | 0.3% (0.11)                    | -2.2%<br>(-0.77)                 | -0.9%<br>(-0.49)               | -0.8%<br>(-0.36) |
|                                  | [0,+20] | 1.0% (0.21)                    | -2.8%<br>(-0.57)                 | 1.6% (0.50)                    | 0.2% (0.05)      |
| Constituent list, 21<br>Aug 2026 | [-1,+1] | 2.0% (1.32)                    | 0.9% (0.51)                      | 1.1% (1.08)                    | 1.1% (0.87)      |
|                                  | [-1,+5] | 0.9% (0.38)                    | -1.6%<br>(-0.63)                 | 0.4% (0.23)                    | 0.2% (0.11)      |
|                                  | [0,+20] | 1.2% (0.30)                    | 0.3% (0.07)                      | 1.8% (0.65)                    | 2.3% (0.65)      |
| Effective date, 21<br>Sep 2026   | [-1,+1] | 0.3% (0.20)                    | 1.4% (1.00)                      | 0.5% (0.46)                    | 0.2% (0.21)      |
|                                  |         |                                |                                  |                                |                  |

*Notes:* CAR = sum of daily portfolio abnormal returns. Portfolio  $t = \text{CAR}/(\sigma\sqrt{L})$ , where  $\sigma$  is the standard deviation of daily portfolio abnormal returns in the estimation window and  $L$  the number of days in the event window; the portfolio approach follows Brown and Warner (1985). The estimation window runs from trading day -130 to -11 and excludes days -1 to +20 around every earlier disclosure (120 days for the announcement, 98 for the confirmation and the constituent list, 89 for the effective date). Market-model parameters are estimated per stock over the same window against the never-named benchmark; the matched benchmark weights controls by their matching weights. Windows are pre-specified and identical for every event; a CAR is called reliable when it is significant at 5% under all four benchmarks. The effective-date windows beyond +1 are not reported because the data end on 23 September 2026. Source: `output/revision2/t3_car_portfolio_clean_est.csv`.

The announcement and confirmation gains behaved differently afterwards. Over days 0 to +20 the announcement CAR lies between -5.5% and 1.6% and is never significant, so the announcement gain faded within a month. Over the same window after the confirmation, constituents earned 6.2% to 10.6%, significant under three of the four benchmarks ( $t$  between 1.94 and 3.03). The fading after the first, conditional announcement is consistent with temporary price pressure (Harris & Gurel, 1986). The persistent gain after the confirmation, once the upgrade was certain and dated, is consistent with a lasting shift in demand for the stocks (Shleifer, 1986; Chen et al., 2004).

Panels B and C separate the stocks that were likely to be included from those

that were eventually passed over. The pre-announcement eligible group earned 3.5% to 7.5% in the announcement week, significant under three of four benchmarks (the matched benchmark gives  $t = 1.91$ ), so the announcement response does not depend on FTSE's later choice. At the confirmation, its three-day CARs of 1.3% to 3.1% are significant under three of four benchmarks but not against matched controls. The named-but-excluded stocks gained 2.3% to 6.2% in the announcement week (significant under three of four benchmarks) and nothing at the confirmation (-1.4% to 0.4%, never significant). No eligibility list was public at the announcement, so the announcement-week gain of stocks that later appeared on the lists suggests that investors responded to characteristics they could observe, such as size and liquidity, rather than to the lists themselves. By the confirmation the preliminary list was public, and only the portfolio of stocks that went on to be included gained.

**Takeaway.** Prices rose in the announcement week and at the confirmation under every benchmark; the confirmation gain persisted under three of four; the constituent list and the first index tranche added no reliable price effect.

#### *4.3 The rebalancing session produced a trading surge only for included stocks*

On 18 September 2026, the last session before the upgrade took effect, constituents' log trading value rose 0.79 above its level on other sessions in the window, relative to never-named stocks (standard error 0.12,  $t = 6.84$ ; Table 5, panel A), and 0.84 relative to matched controls. On the effective date the excess fell to 0.20 ( $p = 0.057$ ) and 0.34 ( $p = 0.007$ ). Named-but-excluded stocks show no surge

on either day (-0.01 and -0.16). Panel B shows mean abnormal trading value on 18 September against each stock's mean from 60 to 6 calendar days before the constituent list: constituents +0.76, matched controls -0.28, named-but-excluded stocks 0.00.

**Table 5. Trading value at the rebalancing session**

Panel A. Excess log trading value on the rebalancing and effective sessions, relative to other sessions from 6 August to 23 September 2026

| Group compared with controls             | 18 Sep 2026 (rebalancing) | 21 Sep 2026 (effective date) | Observations |
|------------------------------------------|---------------------------|------------------------------|--------------|
| Constituents vs never-named stocks       | 0.794*** (0.116)          | 0.202 (0.106)                | 10,820       |
| Constituents vs matched controls         | 0.837*** (0.122)          | 0.341** (0.122)              | 1,536        |
| Named-but-excluded vs never-named stocks | -0.013 (0.116)            | -0.159 (0.136)               | 10,500       |

Panel B. Mean abnormal log trading value on 18 September 2026

| Group              | Mean   | Std. error | Stocks |
|--------------------|--------|------------|--------|
| FTSE constituents  | 0.760  | 0.124      | 24     |
| Matched controls   | -0.277 | 0.102      | 24     |
| Named-but-excluded | 0.000  | 0.153      | 14     |

*Notes:* Panel A: stock and date fixed effects; standard errors clustered by stock. Panel B: abnormal = daily log trading value minus the stock's mean from 60 to 6 calendar days before 21 August 2026. Sources: `output/revision/t4c_spike_test.csv`, `output/revision/t4b_rebalance_groups.csv`. An event regression that uses the last session before 21 August 2026 as the reference gives 0.62 ( $t = 3.54$ ) for 18 September, but that reference session was unusually

active and several earlier sessions then show significant negative coefficients; we therefore compare each session with the average of the window.

This surge matches the trading that funds tracking FTSE benchmarks had to do: they bought the first tranche of constituents at the closing price of 18 September (FTSE Russell, 2026; VIR, 2026), and nothing obliged them to trade the stocks FTSE had named and passed over. Our data do not identify who traded, so we infer index demand from the timing and from the absence of a surge for named-but-excluded stocks. That absence is a direct falsification test: a general rise in trading among large or eligible stocks would have shown up there. The surge moved volume but not prices: the three-day CAR around the effective date is between -0.5% and 0.9% under every benchmark (Table 4), consistent with other investors having bought earlier and supplying the stocks at the close. Because the first tranche carried only 10% of the eventual index weight, this evidence concerns the first tranche; the three later tranches in 2027 may carry larger demand.

**Takeaway.** Trading concentrated at the first-tranche rebalancing close on included stocks only, as index demand would imply, and prices had already adjusted by then.

#### *4.4 The rebalancing surge rises with FTSE size segment*

FTSE assigned the constituents to size segments on 21 August 2026: three large-capitalization stocks (VCB, VIC, VHM), three mid-capitalization stocks (BID, VPB, HPG) and 18 small-capitalization stocks among those in our sample (VIR, 2026). Index weights, and therefore passive purchases, rise with capi-

talization, so the price-pressure channel predicts a larger rebalancing surge for large-capitalization constituents. Foreign allocations follow benchmark weights (Raddatz et al., 2017), which points the same way for the persistent effect.

Table 6 splits the constituent coefficients by segment. The three large-capitalization stocks show the largest fall in illiquidity: -0.61, -1.35 and -2.20 log points across the three windows, compared with -0.20, -0.63 and -0.82 for mid-capitalization stocks and -0.39, -0.85 and -1.00 for small-capitalization stocks. The ordering between mid and small stocks does not follow size, and with three stocks each in the two upper segments clustered inference is unreliable, so we report those estimates without significance marks and read them as descriptive. The rebalancing surge rises monotonically with segment: abnormal log trading value on 18 September 2026 averaged 1.23 for large-capitalization constituents, 0.98 for mid-capitalization and 0.65 for small-capitalization constituents. This ordering is consistent with H4, although the two upper segments contain three stocks each.

**Table 6. Constituent effects by FTSE size segment**

|                                                    | Large (3 stocks) | Mid (3 stocks) | Small (18 stocks) |
|----------------------------------------------------|------------------|----------------|-------------------|
| log Amihud: Announcement                           | -0.607 (0.273)   | -0.204 (0.119) | -0.387*** (0.113) |
| log Amihud: Confirmation                           | -1.350 (0.413)   | -0.629 (0.076) | -0.855*** (0.133) |
| log Amihud: List and rebalancing                   | -2.198 (0.396)   | -0.819 (0.229) | -1.002*** (0.187) |
| log trading value: List and rebalancing            | 2.332 (0.484)    | 0.896 (0.194)  | 1.048*** (0.160)  |
| Abnormal log trading value, 18 Sep 2026 (mean, SE) | 1.231 (0.110)    | 0.976 (0.294)  | 0.645 (0.150)     |

*Notes:* One regression per outcome with stock and week fixed effects and all segment-by-window interactions; comparison group is never-named stocks; standard errors clustered by stock. The last row reports abnormal log trading value relative to each stock's mean from 60 to 6 calendar days before the constituent list. Sources: `output/revision/t5_segments.csv`, `output/table7b_rebalance_by_segment.csv`. Significance marks are omitted for the large and mid segments, which contain three stocks each.

**Takeaway.** The rebalancing trade was largest for the constituents with the largest index weights; the liquidity gain is largest for the three largest stocks but does not rise monotonically across segments, and every segment gained before the first tranche took effect.

#### *4.5 Volatility rose, so the illiquidity decline is conservative*

Constituents became more volatile after the announcement. Their weekly log absolute return rose by 0.34 and 0.21 log points in the announcement and confirmation windows relative to never-named stocks, both significant at 0.1%; the

list-window coefficient of 0.14 is not significant (Table 7, column 1). Because the Amihud ratio divides absolute returns by traded value, rising volatility pushes the ratio up; the measured decline in illiquidity therefore understates the gain in trading capacity. Controlling for the weekly log absolute return enlarges the constituent Amihud coefficients to -0.57, -1.00 and -1.20 (column 3).

Volatility also explains part of the spread result. The Corwin and Schultz (2012) spread did not narrow: it rose by 0.09 to 0.13 percentage points for constituents (Table 2, column 3), and by 0.11 to 0.14 points in the weighted matched sample, from a pre-period mean of 0.57%. With the volatility control, the increase shrinks to 0.06, 0.08 and 0.12 points (Table 7, column 2). The high-low estimator loads on intraday price ranges, so the residual rise may reflect volatility that the weekly control misses rather than a higher cost of trading. We therefore limit our liquidity claims to price impact and trading activity, the dimensions that the Amihud ratio and trading value capture.

**Table 7. Volatility and liquidity**

|                                       | (1) log absolute<br>return | (2) CS spread,<br>controlling for<br>volatility | (3) log Amihud,<br>controlling for<br>volatility |
|---------------------------------------|----------------------------|-------------------------------------------------|--------------------------------------------------|
| Constituent × Announcement            | 0.341*** (0.058)           | 0.0006* (0.0003)                                | -0.574*** (0.102)                                |
| Constituent × Confirmation            | 0.215*** (0.056)           | 0.0008* (0.0003)                                | -1.003*** (0.122)                                |
| Constituent × List and<br>rebalancing | 0.139 (0.086)              | 0.0012* (0.0005)                                | -1.203*** (0.164)                                |
| log absolute return                   |                            | 0.0010***<br>(0.0001)                           | 0.535*** (0.021)                                 |
| Observations                          | 34,369                     | 34,369                                          | 34,369                                           |

*Notes:* Stock and week fixed effects; comparison group is never-named stocks; standard errors clustered by stock. Sources: `output/revision/t6_volatility.csv`, `output/revision/t7_robustness.csv`.

**Takeaway.** Rising trading value, which more than offset rising volatility, drove the fall in illiquidity.

## 5. Robustness and selection

### 5.1 Robustness of the liquidity result

Table 8 collects the robustness checks for the constituent Amihud result. The coefficients remain negative and of similar size in every specification.

**Table 8. Robustness of the constituent Amihud estimates**

| Specification                                                           | Announcement         | Confirmation         | List and rebalancing | Observations |
|-------------------------------------------------------------------------|----------------------|----------------------|----------------------|--------------|
| Baseline (Table 2, column 1)                                            | -0.392***<br>(0.098) | -0.888***<br>(0.124) | -1.129***<br>(0.178) | 34,369       |
| Size-tercile × week FE                                                  | -0.424***<br>(0.108) | -0.786***<br>(0.139) | -0.978***<br>(0.207) | 34,369       |
| Two-way clustering (stock and week)                                     | -0.392***<br>(0.105) | -0.888***<br>(0.130) | -1.129***<br>(0.207) | 34,369       |
| Excluding Vingroup-family stocks                                        | -0.389***<br>(0.100) | -0.863***<br>(0.114) | -1.007***<br>(0.168) | 34,072       |
| Constituent-specific linear trend                                       | -0.382***<br>(0.098) | -0.874***<br>(0.156) | -1.111***<br>(0.225) | 34,369       |
| Weighted matched sample with constituent-specific trend                 | -0.204 (0.138)       | -0.425* (0.199)      | -0.637* (0.262)      | 4,751        |
| Constituent-specific drift after the announcement                       | -0.231* (0.110)      | -0.446**<br>(0.171)  | -0.541* (0.235)      | 34,369       |
| Excluding July to August 2025 from the pre-period                       | -0.354***<br>(0.105) | -0.850***<br>(0.130) | -1.091***<br>(0.183) | 31,212       |
| Controlling for volatility                                              | -0.574***<br>(0.102) | -1.003***<br>(0.122) | -1.203***<br>(0.164) | 34,369       |
| Excluding bank and securities-firm constituents (11 remain)             | -0.439**<br>(0.163)  | -1.042***<br>(0.210) | -1.398***<br>(0.315) | 33,082       |
| ITT group excluding banks and securities firms (19 remain)              | -0.225* (0.095)      | -0.515**<br>(0.163)  | -0.831***<br>(0.223) | 33,874       |
| Wild cluster bootstrap p-value, all controls (352 clusters, 999 draws)  | < 0.001              | < 0.001              | < 0.001              |              |
| Wild cluster bootstrap p-value, unweighted matched sample (48 clusters) | 0.003                | < 0.001              | < 0.001              |              |

| Specification                                                   | Announcement  | Confirmation | List and rebalancing | Observations |
|-----------------------------------------------------------------|---------------|--------------|----------------------|--------------|
| Randomization inference, one-sided p (500 draws, 85-stock pool) | < 0.002       | < 0.002      | < 0.002              |              |
| Placebo: fake event on 7 April 2025 (pre-period only)           | 0.037 (0.104) |              |                      | 18,163       |

*Notes:* Dependent variable log weekly Amihud illiquidity; comparison group is never-named stocks. Standard errors clustered by stock unless stated. Vingroup-family stocks in the sample are VIC, VHM and VRE (VPL listed too late to enter the sample). Wild cluster bootstrap: restricted residuals, Rademacher weights (Cameron et al., 2008). The drift row adds a constituent-specific linear trend in weeks since the announcement (-0.012 per week,  $p = 0.002$ ). Banks are VCB, BID, VPB, HDB, STB, SHB, SSB and MSB among constituents and EIB in the ITT group; securities firms are SSI, VCI, VIX, VND and HCM. Sources: output/revision/t7\_robustness.csv, output/revision2/t\_matched\_weighted.csv, output/revision2/t\_post\_drift.csv, output/revision2/t\_sector.csv, output/revision/t\_wild\_bootstrap.csv, output/revision/t6\_randomization\_inference.csv.

**Pre-announcement months.** The pre-period deviations in Figure 1 are positive, so they work against the estimated declines rather than for them. Dropping July and August 2025, the two months with the largest positive deviations, changes the estimates to -0.35, -0.85 and -1.09. A constituent-specific linear trend fitted over the full sample leaves the estimates almost unchanged, with a trend coeffi-

cient of -0.0002 per week (standard error 0.003). The most demanding specification, the weighted matched sample with a trend, gives -0.20, -0.42 and -0.64; the confirmation and list estimates are significant at 5%, the announcement estimate is not. We therefore rest the announcement-stage conclusion on the full-sample, ITT and trend estimates and treat it as less certain than the later stages.

**Size-related shocks.** Replacing week fixed effects with size-tercile-by-week fixed effects compares constituents only with stocks in the same size tercile each week. The announcement coefficient rises slightly (-0.42), the later two shrink by 12% and 13%, and all remain significant at 0.1%. The predicted-constituent estimates in Table 3, which use only pre-period size, point the same way.

**Trading value.** Trading value behaves differently from the Amihud ratio: a constituent-specific linear trend fitted over the full sample is strong (0.010 log points per week,  $p < 0.001$ ), and trend-adjusted effects fall to 0.28, 0.39 and 0.43 log points, the last significant only at 10%. We describe the trading-value result as an association.

**Inference and influential stocks.** Two-way clustering by stock and week widens standard errors by less than 17%. Excluding the three Vingroup-family stocks changes the estimates by at most 11%. In the unweighted matched sample, with 48 clusters, the wild cluster bootstrap p-values remain at or below 0.003.

**Sector composition and the pre-funding reform.** Banks and securities firms make up 13 of the 24 constituents and 8 of the 27 stocks in the ITT group, and their trading co-moves with market turnover. Dropping them leaves the results intact: the remaining 11 constituents show declines of 0.44, 1.04 and 1.40 log points, and

the remaining 19 ITT stocks 0.23, 0.52 and 0.83 (Table 8). LSEG (2026) lists the removal of pre-funding requirements for foreign institutional investors among the reforms behind Vietnam's upgrade, and that reform applies to every stock foreign investors can buy. The 85 large never-named stocks in the randomization pool, also open to foreign investors within ownership limits, became more liquid by about one fifth as much as constituents in the list window (-0.21 against -1.13). This comparison does not separate the reform from the upgrade for the largest stocks, and we list it as a limitation.

**Placebo.** A fake event on 7 April 2025, estimated on pre-announcement data only, gives a coefficient of 0.04 (standard error 0.10).

## *5.2 Selection: FTSE's lists partly followed rising liquidity*

The 14 stocks FTSE named as eligible but did not include allow a direct test of selection. As a group they became less illiquid than never-named stocks by 0.29, 0.55 and 0.87 log points across the three windows, 62% to 77% of the constituent effects, with the first estimate significant only at 10% (Table 9, panel A). Their gains are concentrated in two stocks, GEE and BSR; for GEE the timing of the gain points to selection, for BSR it does not. Excluding BSR, whose pre-period spans two trading venues, the group estimates are -0.23, -0.46 and -0.73, with only the last significant at 5%.

FTSE screened the April 2026 list on data as of 31 December 2025, which covers the first twelve weeks after the announcement. GEE entered the list in April; relative to never-named stocks, its illiquidity fell by 1.83 log points between the

announcement and 31 December 2025, inside the data window FTSE screened, and by 1.59 by April 2026 (Table 9, panel C). BSR, the other April addition, improved mostly after the cut-off (-0.39 by 31 December, -1.68 by April), and it moved from UPCoM to HOSE only in January 2025, after the data date of the November list (VietnamPlus, 2024), which gives a reason other than liquidity for its later addition. The twelve stocks named in November 2025 changed little: -0.02, -0.19 and -0.30 log points across the three windows, none significant at 5%. At the stock level, PLX, which FTSE dropped in April, and DPM were the only November names to fall by more than 0.5 log points by the April list.

**Table 9. Named-but-excluded stocks and the timing of FTSE's screening**

Panel A. Named-but-excluded (14) and constituents vs never-named stocks

| Window               | Constituents: log Amihud | Named-excluded: log Amihud | Named-excluded: log trading value |
|----------------------|--------------------------|----------------------------|-----------------------------------|
| Announcement         | -0.392*** (0.098)        | -0.293 (0.154)             | 0.589** (0.186)                   |
| Confirmation         | -0.888*** (0.124)        | -0.554* (0.255)            | 0.683* (0.269)                    |
| List and rebalancing | -1.129*** (0.178)        | -0.865** (0.304)           | 0.653 (0.345)                     |

Panel B. log Amihud by FTSE screening window (W1: 7 Oct to 31 Dec 2025, inside the data FTSE screened for the April list; W2: 1 Jan to 6 Apr 2026; W3: from 7 Apr 2026)

| Group                                                         | W1               | W2                | W3                |
|---------------------------------------------------------------|------------------|-------------------|-------------------|
| Constituents (24)                                             | -0.339** (0.110) | -0.440*** (0.100) | -0.932*** (0.129) |
| Named Nov 2025, not included (12)                             | -0.024 (0.092)   | -0.185 (0.127)    | -0.295 (0.168)    |
| Added Apr 2026, not included: GEE, BSR (point estimates only) | -1.15            | -1.67             | -2.51             |

Panel C. Change in mean log Amihud relative to never-named stocks and to the stock's own pre-announcement level

| Stock | First named | W1    | W2    | W3    |
|-------|-------------|-------|-------|-------|
| BSR   | Apr 2026    | -0.39 | -1.68 | -1.81 |
| GEE   | Apr 2026    | -1.83 | -1.59 | -2.98 |
| DGC   | Nov 2025    | 0.11  | 0.18  | 1.20  |
| DIG   | Nov 2025    | -0.01 | 0.04  | 0.19  |
| DPM   | Nov 2025    | -0.21 | -0.86 | -0.74 |
| DXG   | Nov 2025    | -0.02 | -0.12 | -0.32 |
| EIB   | Nov 2025    | 0.22  | -0.24 | -0.61 |
| FRT   | Nov 2025    | -0.02 | -0.10 | -0.10 |
| KBC   | Nov 2025    | 0.31  | 0.04  | 0.03  |
| KDC   | Nov 2025    | 0.66  | 0.46  | 0.10  |
| KDH   | Nov 2025    | -0.33 | -0.16 | -0.47 |

| Stock | First named | W1    | W2    | W3    |
|-------|-------------|-------|-------|-------|
| PDR   | Nov 2025    | -0.25 | -0.19 | -0.50 |
| PLX   | Nov 2025    | -0.37 | -1.12 | -1.04 |
| SAB   | Nov 2025    | -0.17 | 0.02  | -0.11 |

*Notes:* Panels A and B: one regression each with stock and week fixed effects on all 366 stocks; standard errors clustered by stock. Panel C: descriptive. List membership follows Viet Nam News (2025) and The Investor (2026a, 2026b). Sources: `output/revision/t8_named_excluded.csv`, `output/revision/t8c_named_by_stock.csv`. The two-stock row reports point estimates only, because clustered inference is not informative with two stocks. Because named-but-excluded stocks enter with their own window interactions, the constituent coefficients in panel A equal those in Table 2 to three decimals.

This evidence limits how the constituent results can be read. FTSE’s screens rest on investability, and at least one stock became eligible after its liquidity rose during the screening window. The same mechanism can operate among the final constituents, which FTSE chose on June 2026 data, and selection on those data could also favour stocks with large gains on disclosure dates. The ITT group is not exposed to this selection: fixed on 2024 data, it shows illiquidity declines of 24% to 56% and an announcement-week price response significant under three of four benchmarks. The constituent estimates combine the upgrade’s effect with FTSE’s selection; the ITT estimates average the effect over listed stocks that were and were not included. We report both and treat neither as a bound on the other.

**Takeaway.** FTSE’s lists partly followed liquidity, but stocks fixed as likely

constituents before the announcement also gained, so selection does not account for the whole liquidity response.

## **6. Discussion and conclusion**

Vietnam's reclassification brought liquidity and price gains to likely index stocks at FTSE's disclosures, before the first index tranche took effect. Stocks that FTSE had screened as eligible on data from before the announcement became 24% less illiquid within months and 56% less illiquid by the constituent list; the stocks FTSE finally included became 32% and 68% less illiquid. The confirmation added a discrete step, and after it liquidity kept improving gradually. Constituents' prices rose by 3.9% to 7.8% in the announcement week and by 2.7% to 4.5% around the confirmation, and only the confirmation gain lasted, under three of four benchmarks. The first index tranche added a trading surge at the rebalancing close, largest for the largest constituents and absent for excluded stocks, and no price change.

These findings favour the recognition channel over the price-pressure channel as the source of lasting effects. Prices and liquidity moved with FTSE's public statements, and by the first-tranche rebalancing prices had already adjusted. This reading fits developed-market studies that find lasting liquidity gains after inclusion (Hegde & McDermott, 2003) and frontier-market evidence that index effects reflect investor demand rather than liquidity alone (Biktimirov & Afego, 2026). In Vietnam, demand and liquidity moved together, and both moved at disclosure. A frontier-to-emerging upgrade differs from a routine reconstitution because it

changes which benchmark family can hold the stocks, and the information arrives in advance.

The price evidence also shows how the market learned. In the announcement week, before any list was public, stocks that FTSE's screens would later deem eligible gained as a group whether or not FTSE later included them, which suggests that investors used observable size and liquidity to anticipate eligibility. At the confirmation, once the preliminary list was public, only the portfolio of stocks that went on to be included gained.

The segment evidence adds a second margin. The liquidity gain was largest for the three large-capitalization stocks that dominate the index, which is consistent with investors weighting their attention by expected index weight; at the country level, benchmark weights shape fund allocations (Raddatz et al., 2017). With three stocks in that segment, we read this pattern as descriptive. The results also revise the conclusion that an index-level study of the same market reached. Aggregate liquidity of HOSE indices showed no structural break at FTSE watch-list reviews, which could suggest that classification decisions do not matter for liquidity. The stock-level design shows that the 2025 and 2026 decisions mattered for the stocks they concerned, while the broad market, which contains hundreds of stocks outside FTSE's screens, moved much less. In Vietnam's case, the classification event shifted liquidity toward likely index stocks relative to the rest of the market, and index averages diluted that shift; we do not test whether other stocks lost liquidity in absolute terms.

The selection evidence sets a boundary on these conclusions. FTSE's eligi-

bility lists partly followed rising liquidity, and the final constituents were chosen on data from mid-2026. We therefore report constituent and ITT estimates side by side; both show a large response at disclosure, and the ITT estimates do not depend on FTSE's final choice.

The results carry two implications, both drawn from a single event. For regulators, the liquidity and price response arrived with credible, dated announcements rather than with index trading, which is consistent with clear and early communication of reclassification steps bringing the response forward; one event cannot show that earlier communication causes larger gains. For issuers, the gains concentrated in stocks FTSE could hold, and free float and foreign-ownership headroom enter FTSE's screens and index weights; we do not test whether raising them would change an issuer's outcome.

The study has limits. The sample uses stocks listed on 24 September 2026, which omits stocks delisted during the period. Only three trading days follow the effective date, and the first tranche carried 10% of the eventual weight, so the analysis cannot evaluate the three tranches of 2027. The price data do not identify investor type, so we cannot observe whether foreign investors drove the gains. For the largest stocks, the upgrade cannot be fully separated from the concurrent removal of pre-funding requirements for foreign investors, and one named-but-excluded stock, BSR, changed trading venue during the pre-period. The preliminary and April lists come from press reports of FTSE's screening results rather than FTSE documents. Extending the panel through September 2027 and adding foreign-flow data would test whether the constituent effects persist through the remaining

tranches and whether foreign trading drives them.

## Appendix

**Table A.1. Covariate balance before and after matching (constituents vs never-named stocks)**

| Variable                     | Constituent mean | Control mean, all | SMD, all | Control mean, matched | SMD, matched |
|------------------------------|------------------|-------------------|----------|-----------------------|--------------|
| Propensity score             | 0.522            | 0.035             | 1.80     | 0.500                 | 0.08         |
| Pre-period log trading value | 5.170            | 1.585             | 4.06     | 5.132                 | 0.04         |
| Log median price             | 3.228            | 2.725             | 0.66     | 3.061                 | 0.22         |
| Pre-period return volatility | 0.0206           | 0.0207            | -0.03    | 0.0194                | 0.33         |

*Notes:* Nearest-neighbour matching on a logistic propensity score, three never-named controls per constituent, with replacement; 24 distinct control stocks. Pre-period log trading value is the mean daily  $\log(1 + \text{traded value, VND billion})$ . SMD is the standardized mean difference. Log price and volatility remain imbalanced above 0.10, which is why we pair the matched estimates with the size-by-week and trend specifications. Source: `output/revision/tA1_balance.csv`. Table 2, columns 4 to 6, and the matched benchmark in Table 4 use the matching weights.

**Table A.2. Stocks by FTSE list**

| List                                                                         | Stocks                                                                                         |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Preliminary list, Nov 2025 (data as of 31 Dec 2024), later constituents (15) | GEX, HPG, MSN, SHB, SSI, STB, VCB, VCI, VHM, VIC, VIX, VJC, VND, VNM, VRE                      |
| Preliminary list, Nov 2025, not included (12 on HOSE)                        | DGC, DIG, DPM, DXG, EIB, FRT, KBC, KDC, KDH, PDR, PLX, SAB (HUT trades on the Hanoi exchange)  |
| Added to the April 2026 list (data as of 31 Dec 2025)                        | BID, FPT, NVL (later constituents); GEE, BSR (not included); PLX removed                       |
| Constituents not on either eligible list                                     | HCM, HDB, MCH, MSB, SSB, VPB (in sample); TCX, VCK, VPL (listed after Oct 2025, not in sample) |

*Sources:* Viet Nam News (2025); The Investor (2026a, 2026b); FTSE Russell (2026); VIR (2026).

## Declarations

**Data availability.** Daily price and volume data are public and were retrieved through the vnstock library (VCI source). The downloaded daily files and the R code that reproduces all tables and figures (code/analysis.R, code/analysis\_extensions.R, code/analysis\_revision.R, code/analysis\_revision2.R, code/figure1.R, code/figure2.R; R 4.3.3, fixest 0.14.2, MatchIt 4.5.5, data.table 1.14.10) are available from the corresponding author and will be deposited in a public repository on acceptance.

**Ethics.** The study uses public market data and involves no human participants.

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## **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During the preparation of this work the authors used Claude (Anthropic) in order to organise the analysis code, draft text, simulate peer review and verify references against Crossref. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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